

Synthetic Fair-Value Strategy

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1. Introduction

A prediction-market contract is a binary claim that pays \$1.00 if an outcome occurs and \$0.00 if it does not; before settlement it trades between those bounds at whatever price buyers and sellers agree reflects the outcome's probability. Polymarket US launched regulated, real-money contracts of this kind in January 2026, and its coverage of the 2026 FIFA World Cup listed, for each tracked player in each match, three separate contracts on a central limit order book: at least one goal, at least one assist, and at least one goal-or-assist. Each has its own order book and its own update cadence.

These three contracts are not independent instruments. Scoring a goal is one of the ways to record a goal-or-assist, so the compound contract can never be worth less than either single-stat leg:

$$P(\text{goal} \cup \text{assist}) \geq \max(P(\text{goal}), P(\text{assist})).$$

When the compound contract is quoted less actively than the legs that decompose it, its price drifts below that floor. This is a pricing error the market makes against itself, and it can be measured without taking a position. Across 157,595 simultaneously captured three-leg quotes, the compound bid falls below the goals bid in 28.81% of pre-kickoff observations, and for the most-watched players the compound contract carries a 4.0× wider spread and rests 4.8× less total size than the goals leg it can be priced against.

Whether that staleness reverts is a separate question, and it is tested rather than assumed. Across the patience strategy's positions, the gap between the compound contract and its formula fair value narrows between entry and exit 71.4% of the time, against 18.8% under a volatility-matched random walk ($z = 5.04$, exact $p = 0.00003$). A second strategy trades confirmed public events directly: when a goal, a substitution, or a final whistle makes one side of a contract near-certain, a resting order on the other side has not yet requoted, and whoever acts before the book adjusts is

trading a latency gap.

The purpose of this paper is to measure the pricing inefficiency, test whether it is a consistent and statistically significant effect, and evaluate whether either strategy built around it beats the market's own pricing rather than merely trading faster than it. Only the patience strategy does. The speed strategy is profitable and never lost a position it held to settlement, but because it buys contracts already priced near certainty, most of them at the exchange's \$0.99 price ceiling, its win rate is the arithmetic of those prices and not a measurable edge, a distinction Section 5 makes precise.

2. Data Pipeline and Screening Architecture

This analysis relies on a fully automated data pipeline that identifies, validates, and tracks pre-kickoff prop markets while minimizing contamination from fundamentally driven repricing. The objective is a clean sample of simultaneously observed three-leg quotes and a measurement of price behavior in the window where the compound contract de-stales.

The pipeline runs unattended on Windows Task Scheduler and integrates three data sources. First, market and event data is read from Polymarket US's public gateway API, which carries a live score and match clock on each event object. Second, the account's own fills, balances, and settlements are read from the authenticated portfolio API, signed with an Ed25519 key over the message `{timestamp}{method}{path}`. Third, event timestamps used for latency measurement are drawn from neutral match reports (FIFA, ESPN, Al Jazeera), never from Polymarket's own price data, so that the latency figures are not circular.

Polling is tiered by time to kickoff: every twenty minutes beyond six hours out, tightening to every sixty seconds inside the final thirty minutes. Market discovery is structural rather than a hand-maintained list. The set of soccer leagues is pulled live from the platform's own sports endpoint, which returned seventy leagues besides the World Cup as of 2026-08-08, and each league's games are then enumerated against the API.

A clean three-leg observation is recorded only when the goals, assists, and goals-or-assists contracts for the same player can all be read from the same market state in a single pass. Pairing

all three legs at identical capture timestamps yields 157,595 simultaneous observations, of which 124,571 fall in the forty-eight hours before kickoff. This is the sample used throughout the paper.

Every fill the account made is reconciled at ingestion: each trade row's quantity is checked against its cost divided by its price, and each fill is keyed by the platform's own trade identifier, so the local log (879 fills over the window) cannot count one twice. The trading client used for these reads exposes only balance, position, and activity lookups, and contains no order-placement methods.

3. Signal Identification

For a given player in a given match, let g , a , and ga be the prices of the “1+ goals,” “1+ assists,” and “1+ goals-or-assists” contracts. In Sections 4 and 5 each price is the quote midpoint, the average of the best bid and best ask; the 208 entry-time samples below were recorded from last-traded prices. Under a statistical-independence approximation, inclusion–exclusion gives a fair value for the compound contract:

$$P(\text{goal} \cup \text{assist}) = P(\text{goal}) + P(\text{assist}) - P(\text{goal} \cap \text{assist}) \approx g + a - g a.$$

The signal is the difference between that fair value and the contract's actual price:

$$gap = (g + a - g a) - ga.$$

A positive gap means the compound contract is trading cheap relative to its own legs, which is a buy signal. Across 208 clean pre-kickoff samples, gap averaged +0.0106 and exceeded 0.05 in absolute value in 31 of 208 cases. On an instrument priced between \$0.00 and \$1.00, that is a meaningful mispricing.

The reference constraint is the logical floor from Section 1: because a goal is one route to a goal-or-assist, a correctly priced compound contract can never trade below either single-stat leg. A violation of that inequality is a pricing error that needs no model to identify, only three

simultaneous quotes.

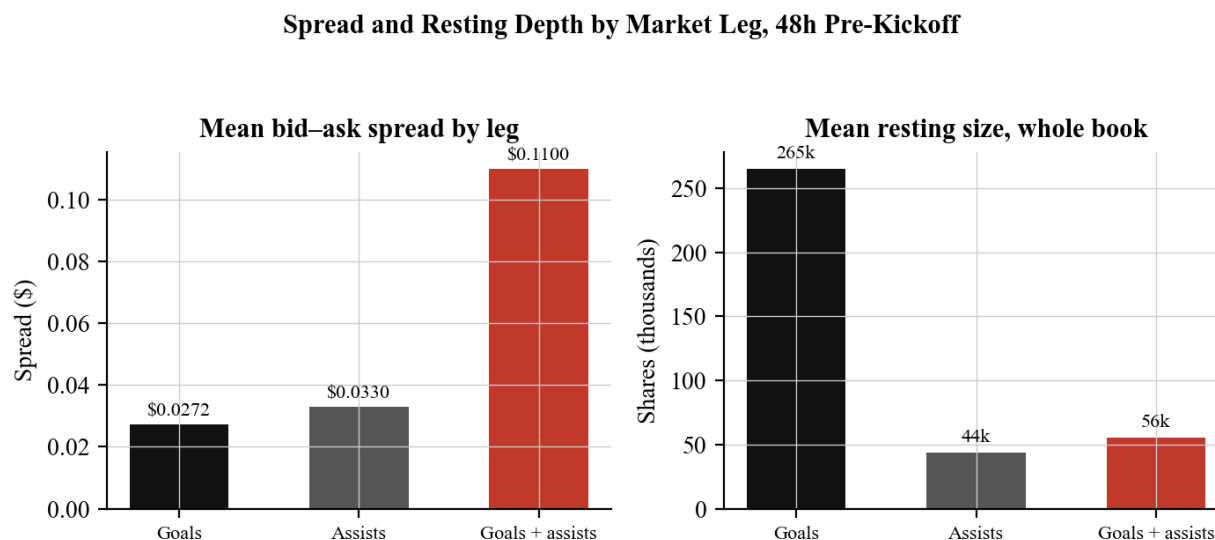
This is not a riskless arbitrage. The independence approximation carries model risk in both directions: a goal is negatively correlated with the same player's own assist, which pushes true $P(\text{goal} \cap \text{assist})$ below $g a$, while a player heavily involved in attack has elevated odds of both outcomes for reasons outside the formula. The strategy is statistical relative value with adverse-selection exposure: a resting bid improved toward fair value can still be filled by a counterparty who is better informed than the formula.

4. Data Analysis

Every snapshot of quotes and book depth is bucketed by time to kickoff. The four figures below test, against the data, the mechanisms the two strategies depend on.

Spread and Resting Depth by Market Leg

The plot below displays the mean bid–ask spread and the mean resting size for each of the three legs, measured over the forty-eight hours before kickoff for the fourteen highest-coverage players.

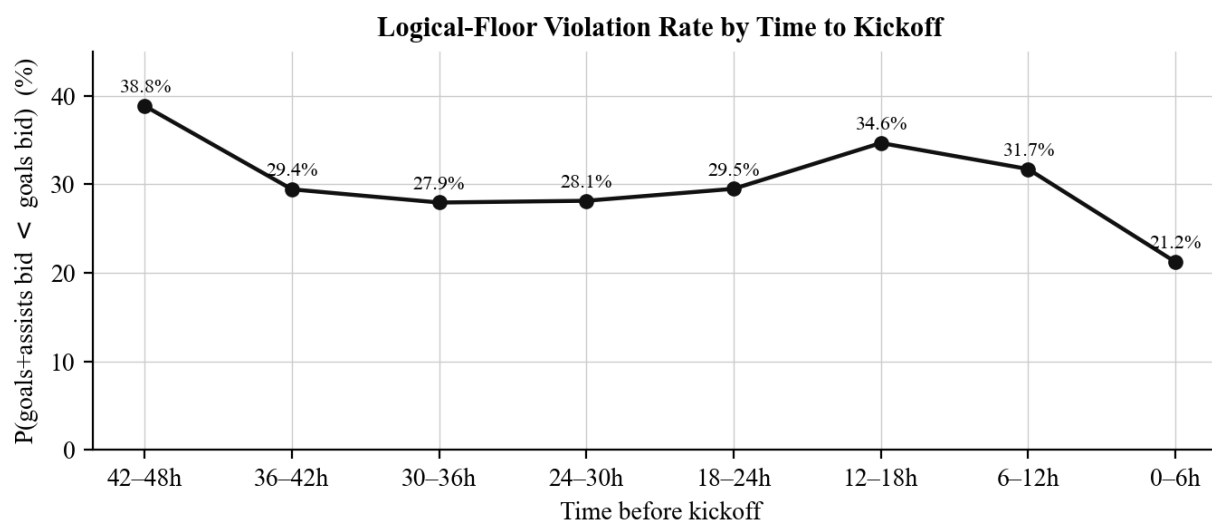


The plot shows that the compound contract is the stale one on both measures at once. It carries a 4.0× wider spread than the goals leg (\$0.110 against \$0.027) and rests 4.8× less total size in its book (55,511 against 265,061 shares), widening to 6.6× in the final four hours. The shortfall

is almost entirely on the offer side: the compound contract's resting bids are as deep as the goals leg's, but far fewer sellers rest behind it. The goals leg is the anchor: its spread stays between \$0.018 and \$0.037 across the two days, and its book grows from roughly 244,000 to 337,000 shares as retail attention arrives. The compound market behaves nothing like it. It opens at a \$0.152 spread and compresses to \$0.050 by kickoff, but its resting size never grows, falling from about 77,000 to 51,000 shares. Retail flow arrives into the simple market and largely ignores the compound one.

Logical-Floor Violation Rate by Time to Kickoff

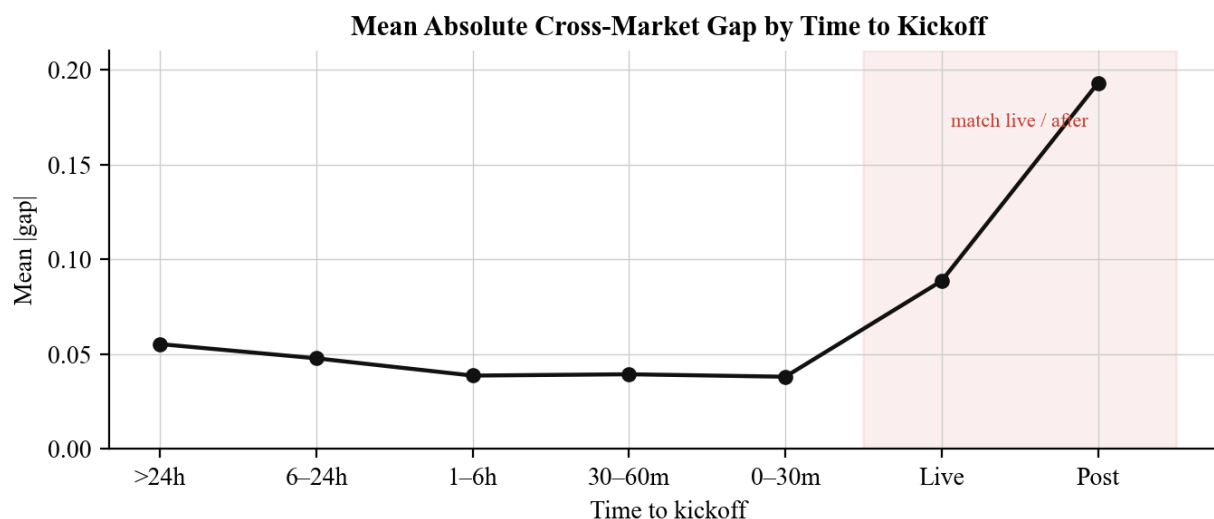
The plot below displays the fraction of pre-kickoff observations in which the compound bid sat below the goals bid, grouped by hours before kickoff.



The plot shows that the best resting bid on the more likely outcome sat below the best resting bid on a subset of it in 28.81% of pre-kickoff observations (35,894 of 124,571), by a mean of \$0.088. The violation rate is highest far from kickoff, 38.83% at forty-two to forty-eight hours out, then holds between 28% and 35% for the next day and a half, and drops to 21.23% only inside the final six hours. The decline is concentrated at the end rather than steady, which is what a de-staling process driven by late retail attention predicts. It also runs opposite to fame. Messi's compound book violated the identity six times in 7,287 observations, while players one tier below the household names, well known enough for their markets to attract flow but not so well known that the compound leg was watched as closely as the simple one, carried violation rates above 20%.

Mean Absolute Cross-Market Gap by Time to Kickoff

The plot below displays the mean absolute difference between the compound price and its formula fair value, by time bucket, from more than a day before kickoff through live play and after the final whistle.

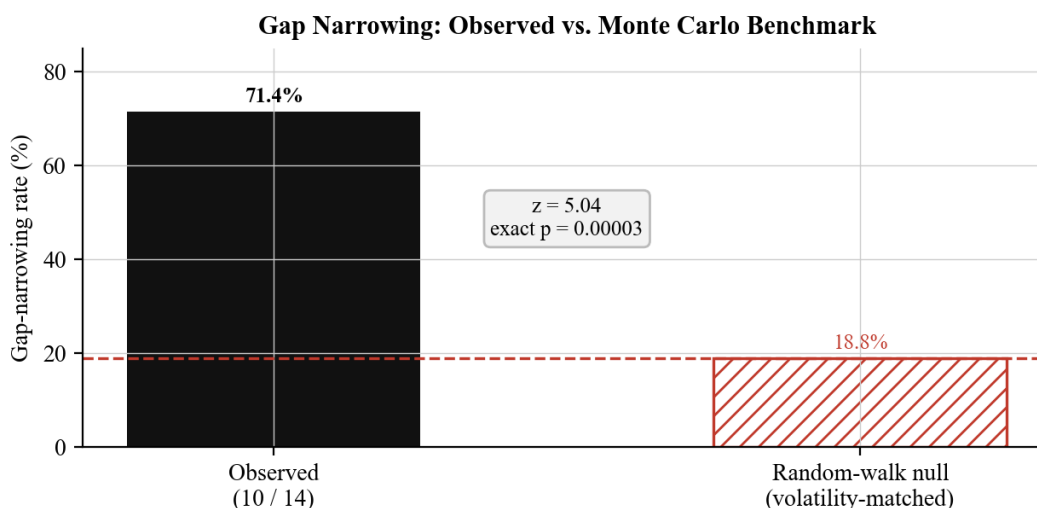


The plot shows that the gap narrows through the pre-kickoff window, from 0.055 more than a day out to about 0.038 in the final hour, and then more than doubles, to 0.089, once the match goes live, rising further after the final whistle as the books empty. Real scoring events hit the three legs at different speeds, and the resulting lag is the inefficiency the second strategy trades.

Gap Narrowing: Observed vs. Monte Carlo Benchmark

To test whether the pre-kickoff mispricing actually reverts rather than merely drifting, per-step increments of *gap* were computed from 147,526 consecutive pre-kickoff same-player, same-game observations (median step about sixty seconds), giving a mean of approximately zero and a standard deviation of $\sigma = 0.0100$ per step. Live-play steps are left out of the calibration: the test scores only pre-kickoff windows, and live steps are roughly three times as volatile, so including them would inflate σ and flatter the test. For each of the fourteen patience-strategy compound positions with a resolvable gap reading at both entry and exit, 5,000 random-walk paths were simulated, each starting at that position's actual entry gap and taking its actual number of observed

steps, and the fraction of paths ending with a smaller absolute gap than they started with was recorded. Exit is the position's own pre-kickoff sale or, where that sale had not filled, kickoff itself, the point at which the strategy is designed to be flat, so every position is scored on the window it was meant to trade whether or not its exit filled in time.



The plot shows that the observed data narrows far more often than the volatility-matched benchmark: 71.4% of the real positions closed toward fair value, against 18.8% of the simulated paths. Because the simulation is calibrated directly from the real minute-by-minute movements and assumes no drift, the difference between 71.4% and 18.8% reflects structure in the price process rather than a parameter choice.

5. Statistical Testing

Each test below is a one-sample proportion z -test. Where the null probability is an entry price rather than a fixed benchmark, the test asks whether the strategy's realized results differ from what the market's own pricing implied. That is the right question for a strategy that might just be fast rather than better informed.

Test 1: Is the gap mean-reversion real?

The null hypothesis is that the gap signal follows a driftless, volatility-matched random walk, which narrows 18.8% of the time.

$$H_0 : p = 0.188 \quad H_1 : p > 0.188 \quad n = 14, \quad \hat{p} = 0.714$$

$$z = \frac{0.714 - 0.188}{\sqrt{\frac{0.188(1 - 0.188)}{14}}} = 5.04, \quad p_{\text{exact}} = 0.00003.$$

At $n = 14$ the normal approximation is loose, so the exact binomial tail is the statement relied on: $P(X \geq 10 \mid n = 14, p = 0.188) \approx 2.6 \times 10^{-5}$, and we reject the null. The result does not depend on the price definition: using last-traded prices instead of midpoints gives 70.6% against 24.4% ($n = 17$, exact $p = 0.00008$). The four positions that fully sold before kickoff, the narrowest possible reading of the strategy, narrowed in three of four. One caveat: contract prices live in $[0, 1]$, so *gap* is bounded and the simulated walk is not, and a bounded process that starts far from zero has to revert eventually because it has nowhere else to go. Two things limit how much of the result that can explain. First, the effect is large: a boundary artifact does not plausibly produce a 71.4% against 18.8% margin at these gap sizes. Second, the effect survives the boundary directly: re-running the null with reflecting barriers at the price bounds changes the simulated rate by less than a point.

Test 2: Is the speed strategy calibrated?

The speed strategy trades a confirmed public event before the resting book requotes. Which side it ends up holding depends on the event: a confirmed goal or assist leaves it long “Yes,” a confirmed substitution or full-time whistle leaves it long “No.” Both are the same trade, acting on public news faster than the book adjusts, applied to opposite events, so they are tested together. The null hypothesis is that the market was already pricing those contracts efficiently, so the true win probability equals the average price the account paid for the side it held.

Before the result: this test cannot validate the strategy at these odds. That has to be

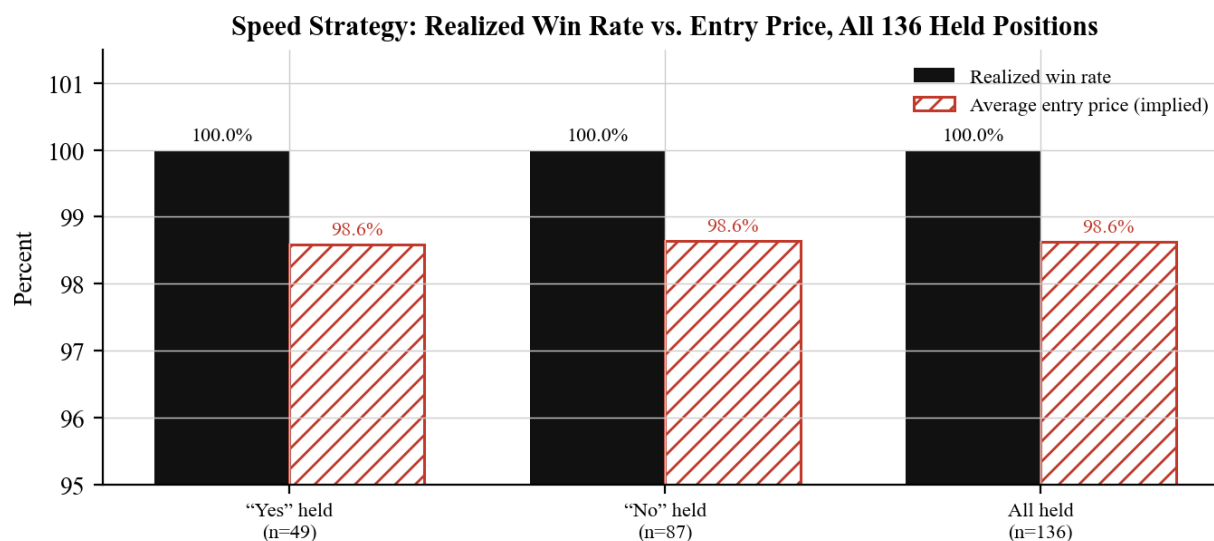
settled before the outcome is read, because it decides what the outcome can mean. Every held position was bought at an average price between \$0.92 and \$0.99, \$0.9862 overall, so the null already expects the strategy to win about 98.6% of the time. With $n = 136$, the best record any strategy can post, winning every single position, gives

$$z_{\max} = \frac{1.000 - 0.9862}{\sqrt{\frac{0.9862(1 - 0.9862)}{136}}} = 1.38, \quad p = 0.168,$$

which does not reach significance, and a one-sided test, which matches the directional claim, gives $p = 0.084$, which does not either. The same holds exactly: under the market's own prices a perfect record has probability $\prod_i p_i \approx 0.15$, well above 0.05. So no outcome this sample could have produced, including a flawless one, would have rejected the null. At these entry prices a perfect record only becomes significant past roughly 216 positions. Whatever the speed strategy's true edge, this test was never able to confirm it here. A failure to reject is therefore the only possible result, and it says nothing about whether the strategy works.

The result.

$$H_0 : p = 0.9863 \quad H_1 : p \neq 0.9863 \quad n = 136, \quad \hat{p} = 1.000, \quad z = 1.38, \quad p = 0.168.$$



The strategy posted that best possible record: all 136 positions carried to a decided settlement resolved the account's way, "Yes" and "No" sides alike, against a 98.6% implied rate. It did not miss once. As established above, the test fails to reject because it cannot do otherwise, not because the strategy fell short.

One fact about the exchange narrows what the profit can mean. Polymarket US contracts trade no higher than \$0.99: no trade or quote in the data exceeds it. 88.7% of the shares the speed strategy bought were bought at exactly that ceiling, and those shares produced \$395 of its \$541. A contract at the ceiling has already been repriced as far as the book allows, so for most of the strategy there was no stale price left to buy. What those positions earned is the final cent to settlement, compensation for holding a near-certain contract to resolution and for the risk that the event is reversed. Only the shares bought below the ceiling, at \$0.81 to \$0.98, which produced the other \$146, could reflect a book that had not yet caught up.

Beyond that, two readings remain open, and the data cannot choose between them. The null assumes that the price the account paid was already the fair probability. If instead the book lagged the event, the record reflects a small timing edge on top of the settlement carry. If the null is right, the account should have expected about two losses in 136, and a result at least this profitable would occur about 15% of the time. The profit is real and mechanically reliable. What separates the two readings is more data, not a different test: roughly 216 positions at these prices, or event-level evidence that entries landed below the post-event fair value.

What the win rates do and do not show

The speed strategy's headline win rate, 100% across its 136 decided held positions, invites a reading it cannot support. The strategy buys contracts the market has already priced between roughly 92% and 99% and holds them to settlement, so a near-perfect win rate is what those entry prices predict rather than a result. A perfectly calibrated market handed the same trades would produce nearly the same number. The question that separates edge from arithmetic is whether the realized rate beats the price paid. Test 2 answers it: the realized rate sits above the price paid, but by less than a point and a half, a margin this sample cannot distinguish from chance. That is

consistent with a strategy that prices better than the market, and equally consistent with one that does not.

That leaves the speed strategy's profit resting on a timing claim: it buys after a goal, substitution, or whistle is public but before the resting book requotes. A win rate cannot establish this, because a stale price and a fair price at 99% resolve the same way. The only direct evidence is the reaction-time sample in Section 6: seven independently timestamped goals, a mean latency of 91 seconds. It shows how fast the account acted, not that it beat the book: all seven entries filled at \$0.99, the ceiling, which is where a fully repriced book sits. Seven observations is a real sample but a thin one, and it is the only evidence the speed strategy has. Only the patience strategy has an edge that survives a test against the market's own pricing.

Multiple comparisons and in-sample validity

Bonferroni-correcting the two tests above (threshold $0.05/2 = 0.025$) changes no conclusion: the mean-reversion test clears it by orders of magnitude, and the speed-strategy calibration test does not reach even the uncorrected 0.05. The bigger problem is that the gap signal was designed and evaluated on the same thirty-five-day dataset, with no out-of-sample holdout. A complete census of every soccer competition Polymarket US lists finds that none of the other seventy leagues currently carries the three-parallel-market structure the signal requires, so a genuine replication needs a second tournament that does not yet exist. The collector already discovers events across all seventy leagues, so that data begins accumulating the moment one lists a player goal or assist market again.

6. Risk

A key risk to the patience strategy arises when the compound contract's staleness does not close before kickoff. The exit is a passive resting sell rather than a market order, so it depends on a counterparty's order arriving. When it does not fill in time, the position is carried into live play, where match variance sits on top of an already-late exit. Of 40 positions, only 13 fully exited before kickoff as intended; the remaining 27 were carried into the match. The 13 clean exits won every

position that did not close flat (12 of 12); the 27 delayed positions won only 84.6%. Their typical return was similar (a mean of 27.1% per position against 28.5%, and a slightly higher median); the difference is in the losses, and all four of the strategy's losing positions were carried into the match. That is the cost of fill risk. The clean-exit number is partly a selection artifact: a position moving the wrong way is more likely to still be open at kickoff than sold at a loss beforehand.

Jeremy Doku's goals-or-assists position ahead of Belgium–Senegal is a clear example. It was bought across three pre-kickoff fills at \$0.29 to \$0.31, a blended \$0.2978, and was not sold before kickoff. Carried into the match, it filled across seven tranches at \$0.38 to \$0.41, realizing +\$25.18 (+27.6%) and +\$23.57 on its two largest tranches. The mispricing it was entered on was real. It closed late, and a first move the other way would have produced the opposite result on the same delay.

The speed strategy carries a different risk, and it is the mirror image of its win rate. Every held position was entered at an average price between \$0.92 and \$0.99, so a winning position earns one to eight cents a share while a position that resolves against the call loses its entire cost. None did: all 136 decided positions resolved the account's way. But the average speed position cost \$319 and returned \$3.92, so a single reversal, a goal overturned on review or a substitution recorded in error, would erase the profit of roughly eighty winning positions. The strategy's record is only as good as the confirmation behind each entry.

To keep the latency measurement honest, event timestamps are taken only from neutral match reports, never from Polymarket's price data, and second-half goals are excluded rather than approximated. Converting a second-half minute to a wall-clock time requires the real halftime-break length, and assuming a standard fifteen minutes can swing a computed latency by 300 seconds, more than the quantity being measured. The seven first-half goals that pass this rule:

Player	Match	Goal	Latency
Bukayo Saka	France–England, 07-18	37'	4.9 s
Nicolas Pépé	Curaçao–Ivory Coast, 06-25	7'	65.7 s
Arda Güler	Türkiye–USA, 06-25	10'	84.3 s
Fabián Ruiz	Spain–Belgium, 07-10	30'	96.8 s
Leandro Trossard	New Zealand–Belgium, 06-26	28'	120.2 s
Thelo Aasgaard	Norway–France, 06-26	21'	125.3 s
Ousmane Dembélé	Norway–France, 06-26	7'	140.2 s

The mean is 91.0 seconds and the median 96.8. Every one of the seven entries filled at \$0.99.

7. Execution Model

Pre-Trade Identification and Filtering

- **Clean three-leg observation.** A candidate surfaces only when all three legs quote from the same pre-kickoff market state and the gap exceeds a threshold; $|gap| > 0.05$ held on 31 of 208 clean samples.
- **Star-player universe.** Positions concentrate on high-coverage names (Messi, Mbappé, Haaland, Kane and similar), whose compound markets de-stale reliably enough to exit into before kickoff. An obscure player can carry the same gap on paper, with far less confidence it closes and far less volume to exit into.

Entry Logic

- **The fair-value ceiling.** Rather than crossing the spread, the patience strategy repeatedly improves its own resting bid in small increments, capped strictly below the formula-implied fair value, so the position buys at or below its own fair-value estimate even in the best case for a counterparty. On the fills, 96.1% of pre-kickoff compound buys with a concurrent quote on both legs (49 of 51) were priced below the formula value computed from those leg asks.

- **Passive only.** Section 4 shows the inefficiency sat almost entirely on the bid side. A crossable, locked arbitrage, buying the compound at its ask and selling the goals leg at its bid, existed in only 0.05% of observations. An aggressive strategy would have found nothing to take.
- **The confirmed-event trigger.** The speed strategy triggers on events, not price levels. On a confirmed goal, assist, substitution, or final whistle for a tracked player, the near-certain side is bought at whatever price is showing and held to \$1.00 settlement.

Risk and Position Management

- **Fixed-dollar sizing.** The design sizes positions as a fixed fraction of available capital rather than a share count, for consistency across contract prices. The account did not hold to it: position cost ranged from under \$1 to \$1,573.
- **Hard cap.** The design caps total portfolio risk at 5% of capital, checked at kickoff, cutting back any position built past that fraction with a below-market sell. The record shows it was not enforced: 20 of 40 patience positions and 108 of 138 speed positions individually cost more than 5% of the \$1,200 contributed.
- **Breadth over size.** The speed strategy spreads capital across every still-unresolved candidate at the moment of the trigger rather than concentrating on one, since the confirming event makes each outcome close to certain rather than just likely.

Exit Strategy

- **Staged sell-off.** Inside the final five minutes before kickoff, the patience strategy sells roughly a quarter of whatever remains once per minute, crossing the spread each round to guarantee that tranche fills.
- **Unconditional backstop.** A final sweep at kickoff closes anything the schedule leaves behind, so the position is meant to be flat by kickoff, a property the trade record found missing in 27 of 40 cases.
- **Hold to settlement.** Once a speed-strategy entry lands near \$0.98 to \$0.99, the remaining margin is too thin, net of the trading fee, to be worth an active sell. Zero of 40 patience-strategy

positions were ever sold at a price at or above \$0.90, and 137 of 138 speed-strategy positions were held to settlement.

The live order-placement layer for the patience strategy was developed jointly with a collaborator on a separate account and is described here as reconstructed from that logic and cross-checked against the surviving trade ledger; every figure in this paper derives from the author's own account and its own records. The sizing and exit rules above are that design. Where the account's own record departs from them, the departure is reported alongside the rule.

8. Final Weights

Two execution schedules follow from the model above. The pre-kickoff exit, which exists to close the fill-risk gap in Section 6, sells a fixed fraction of the remaining position each minute inside the final five, with an unconditional backstop at kickoff:

Minutes to kickoff	Action	Fraction of remainder	Cumulative % closed
5	Staged sell	25%	25.0%
4	Staged sell	25%	43.8%
3	Staged sell	25%	57.8%
2	Staged sell	25%	68.4%
1	Staged sell	25%	76.3%
0	Unconditional sweep	100%	100%

Entries are not scheduled the same way. They are placed as the signal appears, across the full pre-kickoff window. Of 89 pre-kickoff compound buys, the median fill landed 11.7 hours before kickoff (mean 18.6 hours), and the fills weight toward the final six hours rather than spreading evenly:

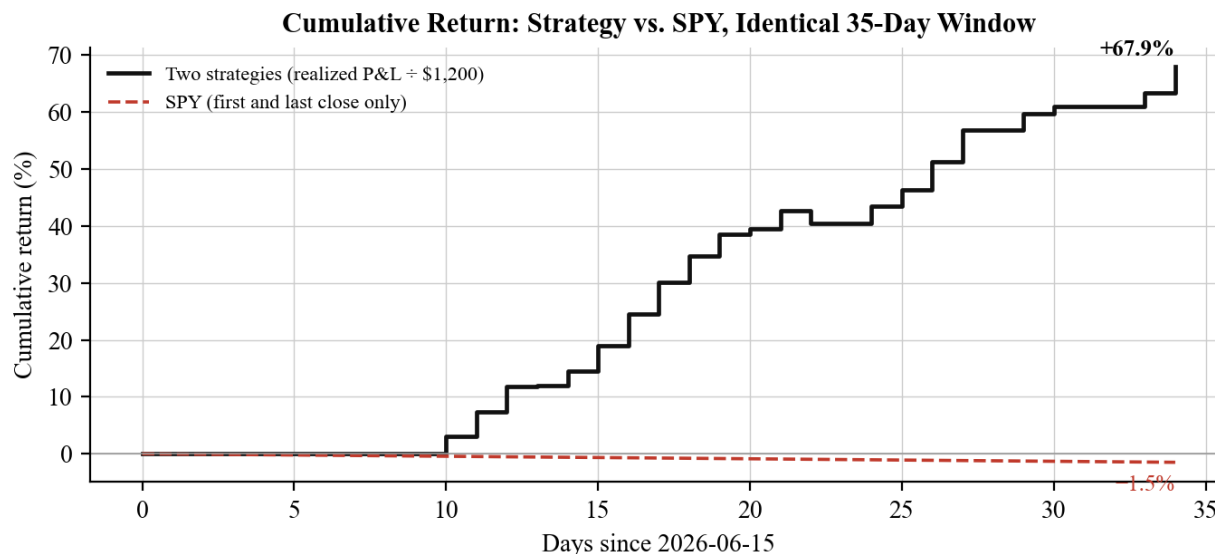
Window before kickoff	Share of pre-kickoff compound buys
More than 24 hours	21%
18–24 hours	19%
6–18 hours	20%
Final 6 hours	39%

9. Results

The strategies were run against Polymarket US's World Cup player-prop markets over a thirty-five-day window, 2026-06-15 to 2026-07-19. Net capital contributed was \$1,200 (four completed deposits totaling \$1,300, less one \$100 withdrawal); the account carries zero open positions at close, so every figure reported is fully realized. Every profit figure is computed from the account's own trade and settlement records and checked against a rebuild from raw fills and settlement payouts, not read from the platform's balance display. The two strategies this paper analyzes realized \$815.02 over the window (the account as a whole, including trades outside both strategies, realized \$718.02), and that profit has two parts.

\$273.75 came from the patience strategy (40 player-prop positions, 35 of them in the compound contract; 35 closed by an active sell and 5 carried to settlement) and \$541.28 from the speed strategy (confirmed-event trades held to settlement, \$541.10 on the 137 positions carried to the final whistle, plus \$0.18 on one closed beforehand). The speed figure matches the raw-fill rebuild to within \$0.16. One patience position resolved in the account's favor but carries no payout in the platform's settlement record; it is counted at \$0, which understates patience by at most \$7.09. Only these two strategies are analyzed, and every test and figure below covers them alone; team markets (exact score, advancement, match winner, spreads) and player shot props are outside both.

Against SPY over the same calendar window (real closes \$754.83 on 2026-06-15, \$743.29 on the last trading day before the window ended), the two analyzed strategies returned +67.9% on contributed capital (\$815.02 on \$1,200), against SPY's −1.5%. The comparison establishes only that the return is not an artifact of a broad-market tailwind.



The two strategies behave differently, and only one carries a statistically validated edge:

	Patience	Speed strategy (held)
Positions	40	137
Win rate	89.5%	100%
Net P&L	+\$273.75	+\$541.10
Validated edge?	Yes ($p_{\text{exact}} = 0.00003$)	Untestable at these odds ($p = 0.168$)

The speed row covers the 137 positions held to settlement (136 decided, one voided), on which the calibration test is run; the strategy realized a further \$0.18 on one position closed beforehand, for a total of \$541.28 as broken out above. Win rates are over positions that did not close flat (34 of 38 for patience, 136 of 136 for speed). The win-rate row is there for completeness but tells you almost nothing about the speed strategy: it bought contracts the market had already priced between 92% and 99%, so a near-perfect win rate is what those entry prices predict rather than a result (Section 5). The “validated edge” row is the one that separates the two. The patience strategy clears it; the speed strategy does not fail it but cannot be tested by it at this sample size, since even its perfect record is not significant at these prices (Section 5).

Within the patience strategy, the fill-risk split is the more informative comparison: the 13

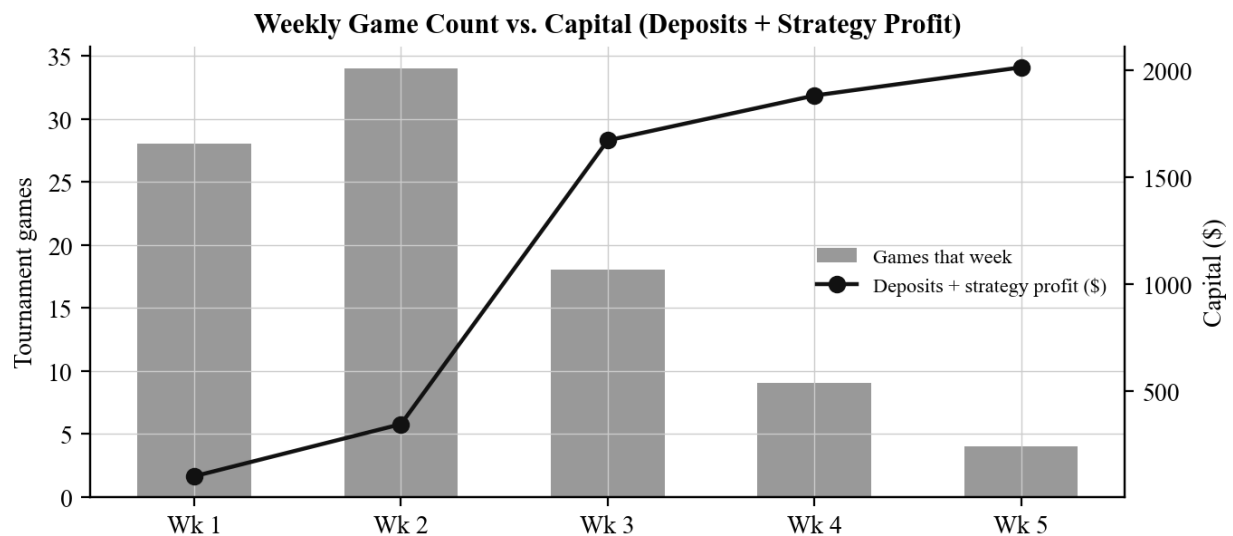
positions that exited before kickoff won all 12 that did not close flat, and the 27 carried into live play won 84.6%.

Metric	Strategy	Benchmark (SPY)
Cumulative return, 35 days	+67.9%	−1.5%
Sharpe ratio, daily ($n = 25$)	1.25	n/a
Maximum drawdown	\$27.81 (5.4% of peak)	n/a

The annualized Sharpe is deliberately not reported. Twenty-five daily observations (2026-06-25, the first day a strategy position closed, through 2026-07-19) from a single bounded event put a wide confidence interval on the daily estimate, and scaling that interval by $\sqrt{365}$ produces a range wide enough to be uninformative while still looking precise. Restricting the calculation to the period after all \$1,200 was deposited (the last deposit settled 2026-06-30) gives a daily Sharpe of 1.16 on $n = 20$, somewhat lower once the low-capital ramp-up is removed.

Capacity

The clearest fact in the data about whether this generalizes is that profit followed opportunity, not capital.



Weekly tournament game count ran 28, 34, 18, 9, 4 as the bracket narrowed (weeks counted from 2026-06-15). Capital, measured as completed deposits plus cumulative strategy profit from

the account's own records rather than the platform's balance display, rose from \$100 to \$2,015. Most of the jump in week 3 is the \$1,000 deposit that settled on 2026-06-30, not performance, so the capital line alone says little about capacity. The informative stretch is weeks 3 to 5, when capital held above \$1,600: weekly strategy profit fell from \$329 to \$210 to \$132 as the game count fell from 18 to 9 to 4. With capital no longer the constraint, profit tracked the number of games available to trade. Depth was not the limit at this size either: the compound contract's best bid rested about 1,200 shares on average for the fourteen most-watched players, against a typical patience position of a few hundred shares. The strategy is bound by opportunity, not by capital.

10. Conclusion

This study demonstrates that a thinly-quoted compound prediction-market contract, priced against the more actively quoted single-stat contracts that decompose it, carries a quantifiable, mechanical pricing inefficiency. Its bid falls below the logical floor $P(\text{goal} \cup \text{assist}) \geq P(\text{goal})$ in 28.81% of 124,571 pre-kickoff observations, and that staleness reverts toward fair value far more often than a volatility-matched random walk predicts: 71.4% against 18.8%, exact $p = 0.00003$. This is the only edge in the paper that survives a test against the market's own pricing. A second strategy, trading confirmed public events directly, is profitable and never lost a held position, but its results are not evidence of skill: it buys contracts already priced at 92% to 99%, most at the exchange's \$0.99 ceiling, and holds them, so its perfect win rate is close to what those prices predict, and a calibration test cannot distinguish it from its own entry pricing (100% realized against 98.6% implied, $p = 0.168$). That is a limit of the sample, not a failed test: at these prices even a perfect record is not significant until roughly 216 positions. Most of its profit is the final cent to settlement on contracts already at the ceiling; the rest is consistent with a genuine timing edge and with favorable variance alike. What profit it makes rests on a timing claim the win rate cannot test and that only seven directly measured event latencies support. The two strategies converted \$1,200 into \$815.02 in fully realized profit over thirty-five days, \$273.75 from patience and \$541.28 from speed (the account as a whole realized \$718.02), outperforming SPY (+67.9% against -1.5%) over the identical window.

The inefficiency is real, but it is bound by opportunity. Once capital stopped being the constraint, weekly profit fell in step with the number of games left in the tournament, and positions were already small next to the book, so more capital would not have bought more trades. Ultimately, the findings suggest that a retail-heavy prediction market during a marquee event prices its simple contracts efficiently and its compound contracts against itself, an asymmetry a faster, better-modeled counterparty can capture at small size but not, on this evidence, at institutional scale.